

How the Proof Works

We start with a smooth, incompressible flow of finite energy and keep its viscosity fixed and positive. The problem is to prove that the flow remains smooth for all time. Its velocity u and pressure p satisfy the three-dimensional Navier–Stokes equation:

$$\underbrace{\partial_t u}_{\text{acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

When strain stretches a vortex, incompressibility makes the material tube narrow as it lengthens. If circulation is retained, the narrowing makes it spin faster. Viscosity spreads rotation into the surrounding fluid, so the tube can narrow while losing circulation. Its vorticity $\omega = \nabla \times u$ satisfies

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The hope is simple: viscosity smooths the fluid faster than nonlinear transport can sharpen it.

A smaller core gives viscosity less distance to act across, but the faster flow can also sharpen it more quickly. Under the Navier–Stokes rescaling with $\lambda > 1$,

$$\begin{aligned} u_\lambda(x, t) &= \lambda u(\lambda x, \lambda^2 t), \\ p_\lambda(x, t) &= \lambda^2 p(\lambda x, \lambda^2 t), \end{aligned}$$

transport and diffusion scale together at fixed ν . Smaller scales give viscosity no automatic advantage.

The critical Sobolev order is $s = \frac{1}{2}$:

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Concentration makes the norm grow above that order and shrink below it; at the critical order it stays invariant. Energy lies below this threshold. The flow can become faster and more concentrated while its total energy falls:

$$E[u_\lambda](t) = \lambda^{-1} E[u](\lambda^2 t), \quad E = \frac{1}{2} \|u\|_2^2.$$

An energy bound can survive the concentration we need to exclude.

Both transport and diffusion act over successively shorter times. If the flow could keep repeating this concentration, those times would permit a Zeno cascade at a finite T_* :

$$\tau_j = \lambda^{-2j} \tau_0, \quad \sum_{j=0}^{\infty} \tau_j < \infty.$$

Scaling allows that possibility; the regularity problem is whether the fluid can actually carry it out.

For a finite-time singularity, velocity would have to blow up. That requires acceleration to blow up, then jerk, and so on, and so on. These changes in velocity are governed by the Navier–Stokes equation. Differentiating it shows what each further temporal order involves:

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u \\ &\quad - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u \\ &\quad - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

Each time derivative splits the transport product and applies a Laplacian to the preceding response. Substituting the first equation into the second exposes $\Delta^2 u$; the next substitution exposes $\Delta^3 u$. Higher temporal orders reach further into the spatial derivatives, with the nonlinear terms carried along:

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

At the critical height, this relation exposes the spatial regularity behind those temporal demands. Write $E_s = \frac{1}{2} \|\Lambda^s u\|_2^2$, where $\Lambda = (-\Delta)^{1/2}$, and let $H = E_{1/2}$. Differentiating E_s and integrating the viscous term by parts moves one spatial derivative onto each factor:

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s,$$

with the transport and pressure contribution

$$\mathcal{P}_s = -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla) u + \nabla p) \rangle_{L^2}.$$

Starting at $s = \frac{1}{2}$, each further time derivative exposes the next spatial energy:

$$\begin{aligned} H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= 4\nu^2 E_{5/2} - 2\nu \mathcal{P}_{3/2} + \partial_t \mathcal{P}_{1/2}. \end{aligned}$$

Continuing through $H''', H^{(4)}, \dots$ reaches $E_{7/2}, E_{9/2}, \dots$, together with the differentiated nonlinear contributions.

Those spatial orders are enough for smoothness. Any chosen number k of continuous spatial derivatives needs only a finite Sobolev height:

$$m > k + \frac{3}{2} \implies H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3).$$

Every choice of k is covered by a sufficiently high half-integer order. With finite energy, their intersection is

$$L^2 \cap \bigcap_{n \geq 0} \dot{H}^{n+1/2} = \bigcap_{m \geq 0} H^m = H^\infty \hookrightarrow C^\infty.$$

The temporal requirements of a singularity have led us into the spatial requirements of smoothness. If these orders remain controlled together through T_* , their limiting velocity is smooth enough to start the flow again.

If a singularity were to form, we would expect the opposite: smoothness at every earlier instant, but an unbounded fixed Sobolev norm as t approaches T_* . The temporal responses would become unbounded too, even while total energy remained finite. Having all finite derivatives before T_* would not give us the control needed to carry them through it.

That control must come from the initial velocity $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$ under the equation. Incompressibility fixes pressure through $-\Delta p = \partial_i \partial_j (u_i u_j)$, with its decay normalization. Writing $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$, the velocity and pressure responses obey

$$\begin{aligned} V_{n+1} &= \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \\ Q_n &= R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}. \end{aligned}$$

Here R_i are the Riesz transforms and repeated spatial indices are summed. Starting with $V_0 = u_0$ at $t = 0$, these relations determine each subsequent response. Spatial differentiation fills out each temporal row. The resulting collection of velocity and pressure derivatives is the mixed jet generated by the datum.

We can retain any finite number of temporal rows and spatial orders. Choosing a temporal depth N and spatial height M gives a rectangle $\mathcal{R}_{N,M}[u_0; T_*]$. The datum-generated rectangle theorem, Theorem 4.6, controls its adjacent temporal rows throughout the approach to T_* :

$$\sup_{0 < T < T_*} \sum_{n=0}^N \left(\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right) < \infty.$$

The bound can depend on the orders N, M , but remains finite as T approaches T_* . Since $V_{n+1} = \partial_t V_n$, the adjacent rows control both a response and its change in time. The spaces H^{M+1} and H^{M-1} pair across H^M ; their time-integrable product gives a continuous endpoint trace by the Hilbert-triple theorem:

$$V_n \in C([0, T_*]; H^M).$$

A larger rectangle retains the derivatives already present. On each overlap, both rectangles contain derivatives of the original velocity and pressure, so their entries and endpoint traces agree. We can increase temporal depth and spatial height independently without losing that agreement. Their compatible intersection gives

$$\begin{aligned} \mathcal{I}[u_0] &:= (\mathcal{R}_{N,M}[u_0; T_*])_{N,M \geq 0}, \\ u &\in \bigcap_{r,m \geq 0} H^r(0, T_*; H^m(\mathbb{R}^3)). \end{aligned}$$

Every finite request for derivatives fits inside one of these rectangles. In particular, the velocity reaches T_* in every spatial Sobolev space, and its traces agree as one limiting velocity:

$$\begin{aligned} u_* &= \lim_{t \uparrow T_*} u(t) \quad \text{in every } H^m, \\ u_* &\in H_\sigma^\infty \hookrightarrow C^\infty. \end{aligned}$$

Smooth local existence starts the flow again from u_* , and uniqueness joins it to the original solution. It continues beyond its supposed maximal time. A finite T_* is impossible, so $T_* = \infty$.

The constructed intersection now supplies finite estimates at any chosen orders. For any finite time T , temporal order r , spatial order k , and $2 \leq p \leq \infty$, choose $m > k + \frac{3}{2} - \frac{3}{p}$. Sobolev embedding gives

$$\partial_t^r u \in C([0, T]; H^m) \hookrightarrow L^\infty(0, T; W^{k,p}).$$

In particular, taking $r = 0$, $k = 1$, $p = \infty$, and $m = 3$ controls the total strain over any finite time:

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq T \sup_{0 \leq t \leq T} \|\nabla u(t)\|_\infty < \infty.$$

For a locally circular material tube with axial strain σ , $\dot{r}/r = -\sigma/2$ and $\sigma \leq \|\nabla u\|_\infty$. Hence

$$r(t) \geq r(0) \exp\left(-\frac{1}{2} \int_0^t \|\nabla u(s)\|_\infty ds\right) > 0.$$

Only a finite amount of stretching can accumulate in a finite time. The tube cannot be squeezed to zero radius there.